

SIMATS ENGINEERING

CO1 - ASSESSMENT TOOL 1

Analytical Problem Solving

COURSE NAME: DEEP LEARNING

COURSE CODE: MLA0408

FACULTY : Dr. D.SUDHAGAR

COURSE OUTCOME COVERED

CO 1: Learning Algorithms, Maximum Likelihood Estimation (MLE), Building Machine Learning Algorithms, Neural Networks, Artificial Neurons, Perceptron, Multilayer Perceptron (MLP), Forward Propagation, Backpropagation Algorithm, Gradient Descent, Stochastic Gradient Descent (SGD), Mini-Batch Gradient Descent, Curse of Dimensionality. (BTL-4)

CO1 Weightage: 15%

Assessment Tool Weightage: 35%

Duration: 30 minutes

Marks: 25

Code of Conduct:

I Haresh Kumar N L (192425009) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: haresh

QUESTIONS:05

Total Marks:25

Analytical Problem Solving

1. Learning Algorithms (Optimization Behavior)

A machine learning model is trained using gradient descent on a convex loss function. Initially, the learning rate is set very high, and later it is gradually decreased.

Question:

Analyze how the choice of a high initial learning rate followed by decay affects convergence. Under what conditions can this strategy outperform a constant learning rate? Discuss possible risks and justify with reasoning.

2. Maximum Likelihood Estimation (MLE)

Suppose you are given a dataset generated from a Gaussian distribution with unknown mean μ and known variance σ^2

Question:

Derive the Maximum Likelihood Estimator for μ , and analytically explain how the estimate changes when:

- The dataset contains outliers
 - The sample size increases
- What does this imply about the robustness of MLE?

3. Building a Machine Learning Algorithm (Model Selection)

You are designing a classification algorithm for a dataset with 10,000 features but only 500 training samples.

Question:

Analyze the challenges involved in training such a model. Propose a strategy (algorithm pipeline) to handle this situation and justify each step (e.g., feature selection, regularization, dimensionality reduction). Explain how your approach mitigates overfitting.

4. Neural Networks (Multilayer Perceptron - MLP)

Consider a Multilayer Perceptron with one hidden layer using sigmoid activation functions.

Question:

Explain analytically why adding more hidden neurons increases the representational power of the network. Then, reason why this does not always lead to better performance. Support your answer using concepts like overfitting and generalization.

5. Backpropagation, SGD & Curse of Dimensionality

A deep neural network is trained using Stochastic Gradient Descent (SGD) on a very high-dimensional dataset.

Question:

Analyze how the **curse of dimensionality** impacts:

- Gradient updates in backpropagation
- Convergence speed of SGD
- Model generalization

Propose at least two techniques to overcome these issues and justify them analytically.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Answer

1) Gradient Descent updates the model parameters as:

$$\theta_{t+1} = \theta_t - \eta_t \nabla J(\theta_t)$$

where:

- θ = model parameters
- η_t = learning rate
- $J(\theta)$ = loss function

Effect of High Initial Learning Rate

A **high initial learning rate** allows the optimizer to make large parameter updates.

Advantages:

- Reaches the region near the global optimum much faster.
- Escapes flat regions or shallow local minima.
- Reduces initial training time.

Effect of Learning Rate Decay

As training progresses, the learning rate is gradually reduced.

Advantages:

- Enables smaller, more precise updates.
- Prevents oscillation around the minimum.
- Improves convergence stability.

When It Outperforms a Constant Learning Rate

This strategy performs better when:

- The loss function is smooth and convex.
- The initial learning rate is not excessively large.
- Learning rate decay is scheduled properly.
- Training involves many iterations.

A constant learning rate may:

- Converge slowly if too small.
- Oscillate or diverge if too large.

Possible Risks

- Extremely high learning rate may overshoot the optimum.
- Training may become unstable or diverge.
- Too rapid decay may stop learning before reaching the optimum.

Conclusion

Using a **high initial learning rate followed by gradual decay** combines fast exploration with stable convergence, often achieving faster and more accurate optimization than a constant learning rate when tuned appropriately.

2) Assume

$$X_i \sim N(\mu, \sigma^2)$$

where:

- Unknown mean = μ
- Known variance = σ^2

The likelihood function is

$$L(\mu) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x_i - \mu)^2}{2\sigma^2}}$$

Taking logarithm,

$$\log L(\mu) = -\frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2$$

Differentiate with respect to μ :

$$\frac{d \log L}{d\mu} = \frac{1}{\sigma^2} \sum_{i=1}^n (x_i - \mu)$$

Set equal to zero:

$$\sum_{i=1}^n (x_i - \mu) = 0$$

Therefore,

$$\hat{\mu}_{MLE} = \frac{1}{n} \sum_{i=1}^n x_i$$

Effect of Outliers

Outliers greatly affect the sample mean.

- One extreme observation shifts the estimate.
- MLE becomes biased toward the outlier.

- Estimation accuracy decreases.

Effect of Increasing Sample Size

As sample size increases:

- Random error decreases.
- Estimate approaches the true mean.
- Variance of estimator decreases.

Hence,

$$\hat{\mu}_{MLE} \rightarrow \mu$$

(consistency property)

Robustness of MLE

MLE using the sample mean is:

- Efficient for Gaussian data.
- Consistent.
- Not robust to outliers because every observation contributes equally.

3) Given:

- Features = 10,000
- Training samples = 500

This is a **high-dimensional, low-sample-size** problem.

Challenges

- Curse of dimensionality.
- Severe overfitting.
- High computational cost.
- Irrelevant and redundant features.
- Poor generalization.

Proposed Pipeline

Step 1: Data Preprocessing

- Remove missing values.
- Normalize features.

Reason: Ensures stable optimization.

Step 2: Feature Selection

Methods:

- Chi-square

- Mutual Information
- LASSO

Reduce features from 10,000 to a few hundred.

Reason: Removes irrelevant variables and lowers variance.

Step 3: Dimensionality Reduction

Apply PCA.

Reason:

- Removes correlated features.
- Preserves most variance.
- Reduces computational complexity.

Step 4: Regularization

Use:

- L1 Regularization
- L2 Regularization

Reason:

- Prevents large weights.
- Reduces model complexity.
- Improves generalization.

Step 5: Train Classifier

Suitable choices:

- Support Vector Machine
- Logistic Regression
- Random Forest (carefully tuned)

Step 6: Cross Validation

Use k-fold cross-validation.

Reason:

- Better utilization of limited samples.
- Reliable performance estimation.

How Overfitting is Reduced

- Feature selection removes noise.
- PCA lowers dimensionality.
- Cross-validation prevents optimistic bias.

Thus, the model generalizes better to unseen data.

4) A Multilayer Perceptron computes

$$y = f\left(\sum w_i x_i + b\right)$$

where sigmoid activation is

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

Why More Hidden Neurons Increase Representational Power

Each hidden neuron learns a different nonlinear transformation.

Adding neurons allows the network to:

- Learn more complex decision boundaries.
- Represent nonlinear relationships.
- Approximate arbitrary continuous functions (Universal Approximation Theorem).

Hence, model capacity increases.

Why More Neurons Do Not Always Improve Performance

More neurons increase:

- Number of parameters.
- Training complexity.
- Risk of memorizing training data.

This leads to **overfitting**.

The model performs very well on training data but poorly on unseen data.

Generalization

Good generalization means performing well on unseen samples.

Too many neurons reduce generalization because the network fits noise instead of meaningful patterns.

Conclusion

Adding hidden neurons improves representational ability but excessive neurons increase overfitting. An optimal network balances model complexity and generalization.

5) High-dimensional data introduces several learning challenges.

(a) Impact on Gradient Updates

In high dimensions:

- Gradients become noisy.
- Many parameters receive very small updates.
- Vanishing or exploding gradients become more likely.
- Optimization becomes difficult.

(b) Impact on SGD Convergence

SGD uses one sample (or mini-batch) per update.

High-dimensional data causes:

- Higher variance in gradient estimates.
- Slower convergence.
- More oscillations.
- Increased computational cost.

(c) Impact on Generalization

High-dimensional feature spaces increase:

- Model complexity.
- Risk of overfitting.
- Sparse data distribution.

The network may memorize training examples rather than learn general patterns.